

Mark Scheme — Practice Paper 2 (P1)

M = method mark, A = accuracy mark, B = independent mark. 教师/批改用。

1 General term $\binom{6}{k}(x^2)^{6-k}\left(\frac{2}{x}\right)^k = \binom{6}{k}2^k x^{12-3k}$ (M1). Independent of x : $12 - 3k = 0 \Rightarrow k = 4$ (A1).
Term = $\binom{6}{4} \cdot 2^4 = 15 \times 16 = 240$ (A1). [3]

2 $x^2 + 2x + 1 = kx - 3 \Rightarrow x^2 + (2 - k)x + 4 = 0$ (M1). No intersection $\Rightarrow b^2 - 4ac < 0$: $(2 - k)^2 - 16 < 0$ (M1 A1). $\Rightarrow -4 < 2 - k < 4 \Rightarrow -2 < k < 6$ (A1). [4]

3 $2r + r\theta = 20$ and $\frac{1}{2}r^2\theta = 16$ (B1 both). Eliminate θ : $r\theta = \frac{32}{r} \Rightarrow 2r + \frac{32}{r} = 20 \Rightarrow r^2 - 10r + 16 = 0$ (M1). $r = 2$ or $r = 8$ (A1). $r = 2 \Rightarrow \theta = 8 > 2\pi$, impossible, so $r = 8$, $\theta = 0.5$ (A1, with reason). [4]

4(a) Gradient $AB = \frac{5-1}{4+2} = \frac{2}{3}$, gradient $BC = \frac{2-5}{6-4} = -\frac{3}{2}$ (M1). Product = -1 , so $AB \perp BC$ (A1).
[Alternative: Pythagoras with $AB^2 = 52$, $BC^2 = 13$, $AC^2 = 65$.] [2]

4(b) $D = A + C - B = (-2 + 6 - 4, 1 + 2 - 5) = (0, -2)$ (B1). [1]

4(c) $AB = \sqrt{52}$, $BC = \sqrt{13}$ (M1). Area = $\sqrt{52 \times 13} = \sqrt{676} = 26$ (A1). [2]

5(a) Common denominator: $\frac{\sin^2 \theta + (1 + \cos \theta)^2}{\sin \theta(1 + \cos \theta)}$ (M1) = $\frac{\sin^2 \theta + 1 + 2 \cos \theta + \cos^2 \theta}{\sin \theta(1 + \cos \theta)} =$
 $\frac{2 + 2 \cos \theta}{\sin \theta(1 + \cos \theta)}$ (M1 using $\sin^2 + \cos^2 = 1$) = $\frac{2(1 + \cos \theta)}{\sin \theta(1 + \cos \theta)} = \frac{2}{\sin \theta}$ (A1 AG). [3]

5(b) $\frac{2}{\sin \theta} = 3 \Rightarrow \sin \theta = \frac{2}{3}$ (M1). $\theta = 0.730$ (A1), $\theta = \pi - 0.7297 = 2.41$ (A1). Answers in degrees (41.8°, 138.2°) lose the final A mark. [3]

6(a) $f(x) \leq 4$ (B1). [1]

6(b) $y = 4 - x^2 \Rightarrow x^2 = 4 - y \Rightarrow x = \pm\sqrt{4 - y}$ (M1). Since $x \leq 0$: $f^{-1}(x) = -\sqrt{4 - x}$ (A1).

Domain $x \leq 4$ (B1). [3]

6(c) $gf(x) = 2(4 - x^2) + 1 = 9 - 2x^2$ (B1). $9 - 2x^2 = -9 \Rightarrow x^2 = 9$ (M1). $x = -3$ only (A1; $x = 3$ not in domain of f). [3]

7(a) $\frac{dy}{dx} = 3(2x - 3)^2 \times 2 - 6 = 6(2x - 3)^2 - 6$ (M1 chain rule, A1). $\frac{d^2y}{dx^2} = 12(2x - 3) \times 2 =$
 $24(2x - 3)$ (A1 FT). [3]

7(b) $6(2x - 3)^2 = 6 \Rightarrow (2x - 3)^2 = 1 \Rightarrow 2x - 3 = \pm 1$ (M1). $x = 1$: $y = (-1)^3 - 6 = -7$; $x = 2$:
 $y = 1 - 12 = -11$ (A1 both points (1, -7), (2, -11)). At $x = 1$: $\frac{d^2y}{dx^2} = -24 < 0$, maximum (A1). At
 $x = 2$: $\frac{d^2y}{dx^2} = 24 > 0$, minimum (A1). [4]

8(a) $\frac{dV}{dr} = 4\pi r^2$ (B1). $\frac{dr}{dt} = \frac{dV}{dt} \bigg/ \frac{dV}{dr} = \frac{12}{4\pi(2)^2}$ (M1) = $\frac{3}{4\pi} = 0.239 \text{ cm s}^{-1}$ (A1). [3]

8(b) $\frac{dy}{dx} = 2x - 3 = 5$ (M1) $\Rightarrow x = 4$, $y = 16 - 12 + 5 = 9$: point (4, 9) (A1). Tangent: $y - 9 =$
 $5(x - 4)$ (M1) $\Rightarrow y = 5x - 11$ (A1). [4]

9(a) $S_1 = a = 5$ (B1). $S_2 = 16 \Rightarrow u_2 = 11$ (M1) $\Rightarrow d = 6$ (A1). $u_{20} = 5 + 19 \times 6 = 119$ (A1). [Also
accept $u_n = S_n - S_{n-1} = 6n - 1$.] [4]

9(b)(i) $(x - 2)^2 = (x + 4)(x - 5)$ (M1) $\Rightarrow x^2 - 4x + 4 = x^2 - x - 20 \Rightarrow x = 8$ (A1). [2]

9(b)(ii) Terms 12, 6, 3; $r = \frac{1}{2}$ (B1 FT). $S_{\infty} = \frac{12}{1 - \frac{1}{2}} = 24$ (B1). [2]

10(a) $9 - x^2 = 5 \Rightarrow x = \pm 2$ (B1). Area $= \int_{-2}^2 ((9 - x^2) - 5) dx = \int_{-2}^2 (4 - x^2) dx$ (M1) $= [4x - \frac{1}{3}x^3]_{-2}^2$ (A1) $= (8 - \frac{8}{3}) - (-8 + \frac{8}{3}) = \frac{32}{3} = 10.7$ (A1). [Symmetry: $2 \int_0^2$ acceptable.] [4]

10(b) Rotation about the y -axis: $V = \pi \int x^2 dy$ with $x^2 = 9 - y$ (M1). Limits $y = 0$ to $y = 9$ (B1). $V = \pi \int_0^9 (9 - y) dy = \pi [9y - \frac{1}{2}y^2]_0^9$ (M1) $= \pi(81 - 40.5) = 40.5\pi$ (A1; $= \frac{81}{2}\pi$). [4]

11(a) $r^2 = (5 - 2)^2 + (3 + 1)^2 = 25$ (M1). $(x - 2)^2 + (y + 1)^2 = 25$ (A1). [2]

11(b) $x = 10 - 2y$ substituted: $(8 - 2y)^2 + (y + 1)^2 = 25$ (M1) $\Rightarrow 5y^2 - 30y + 40 = 0 \Rightarrow y^2 - 6y + 8 = 0$ (A1) $\Rightarrow y = 2, 4$; $P(6, 2)$, $Q(2, 4)$ (A1, either order). [Substituting $y = \frac{10-x}{2}$ gives $x^2 - 8x + 12 = 0$.] [3]

11(c) Gradient of radius $= \frac{2 - (-1)}{6 - 2} = \frac{3}{4}$ (M1 A1). Tangent gradient $= -\frac{4}{3}$ (M1). $y - 2 = -\frac{4}{3}(x - 6) \Rightarrow 4x + 3y = 30$ (A1). [4]

12(a) Let $u = \sqrt{x}$: $u^2 - 5u + 6 = 0 \Rightarrow (u - 2)(u - 3) = 0$ (M1 A1). $\sqrt{x} = 2$ or $3 \Rightarrow x = 4$ or $x = 9$ (A1). [3]

12(b) $b^2 - 4ac > 0$: $16 - 4k^2 > 0$ (M1 A1) $\Rightarrow k^2 < 4 \Rightarrow -2 < k < 2$ (A1). Also $k \neq 0$ (otherwise the equation is linear with one root) (B1). Answer: $-2 < k < 0$ or $0 < k < 2$. [4]

Total: 75